

Instance Segmentation

2018

Mask R-CNN

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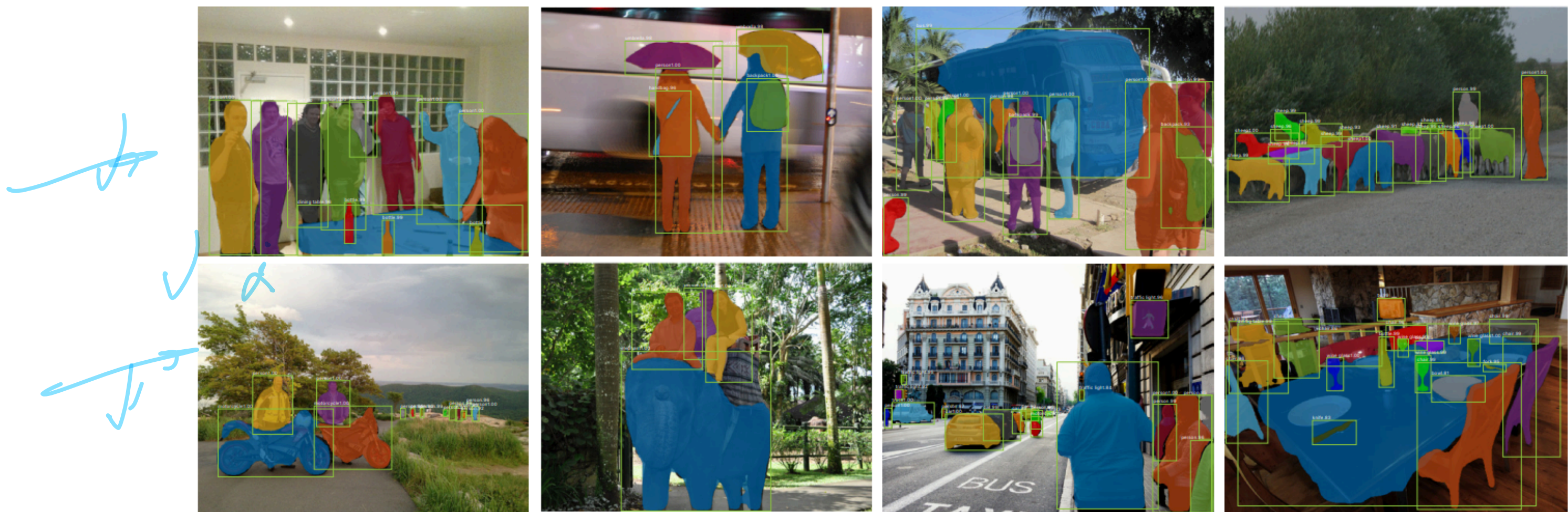


Figure 2. **Mask R-CNN** results on the COCO test set. These results are based on ResNet-101 [19], achieving a *mask* AP of 35.7 and running at 5 fps. Masks are shown in color, and bounding box, category, and confidences are also shown.



Figure 4. More results of **Mask R-CNN** on COCO test images, using ResNet-101-FPN and running at 5 fps, with 35.7 mask AP (Table 1).

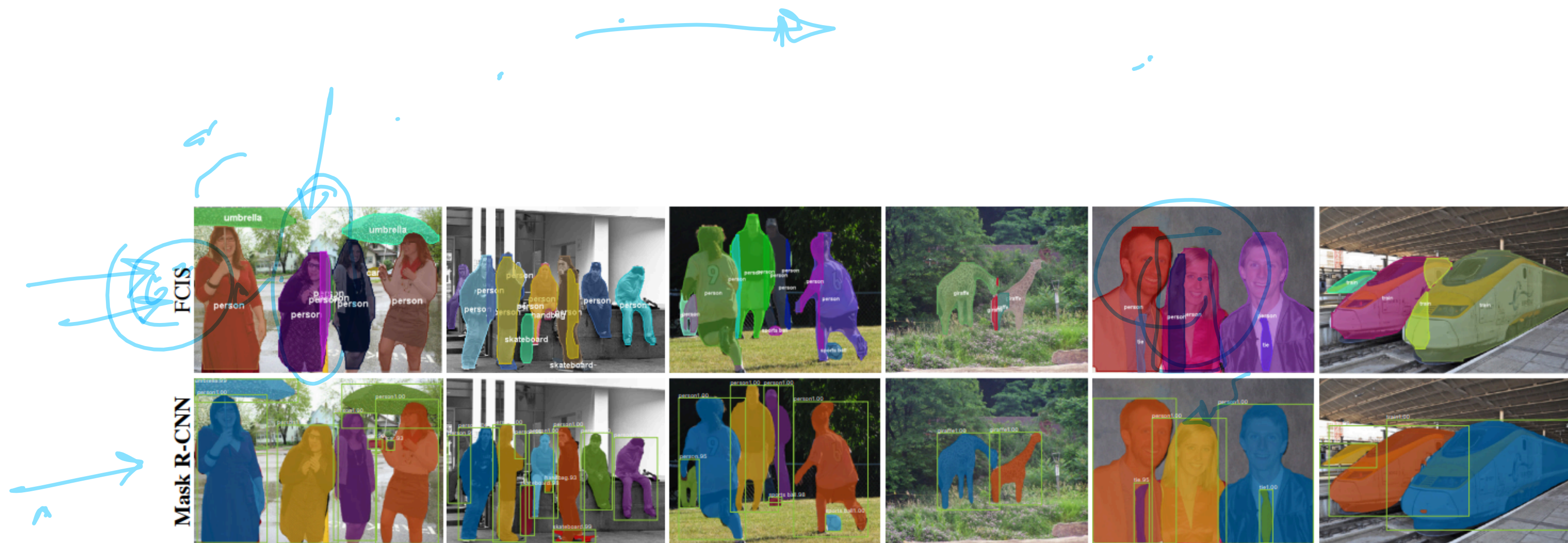


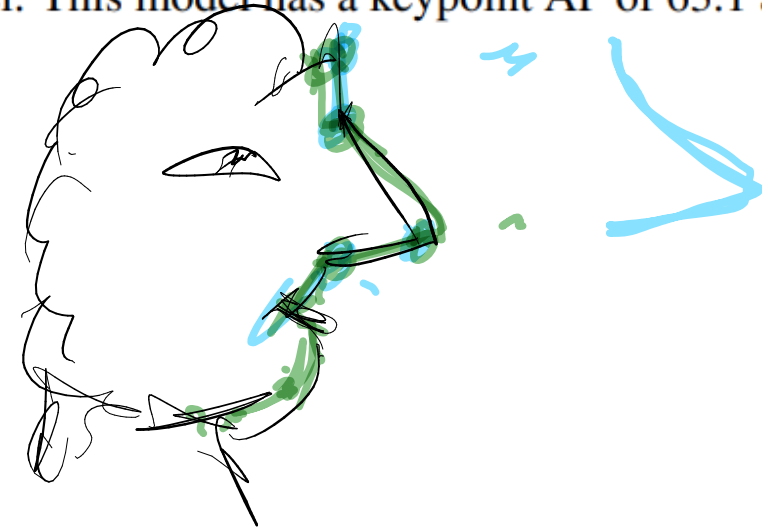
Figure 5. FCIS+++ [26] (top) vs. Mask R-CNN (bottom, ResNet-101-FPN). FCIS exhibits systematic artifacts on overlapping objects.

Keypoint Det! → { → body seg. → action recog./medical!

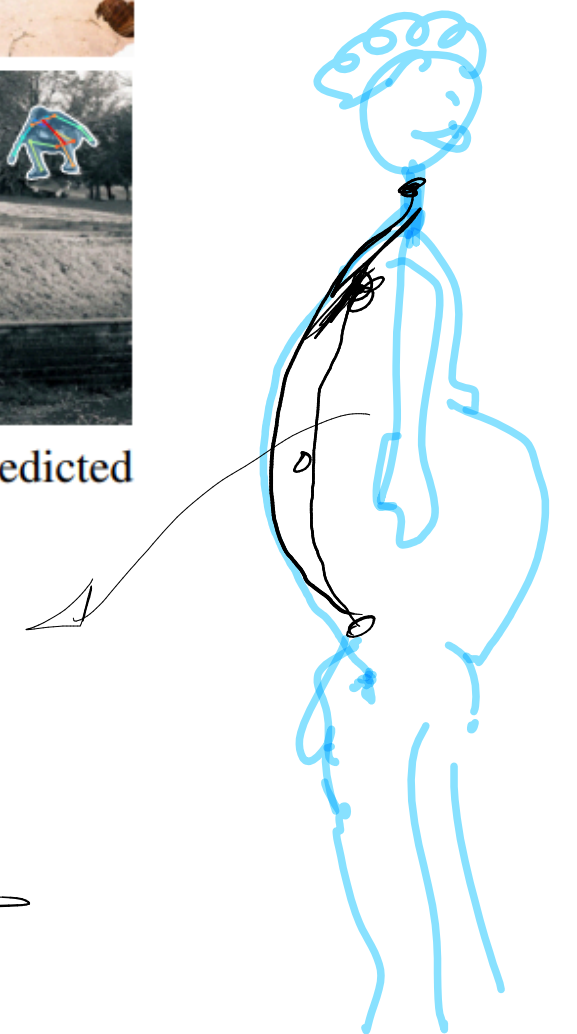


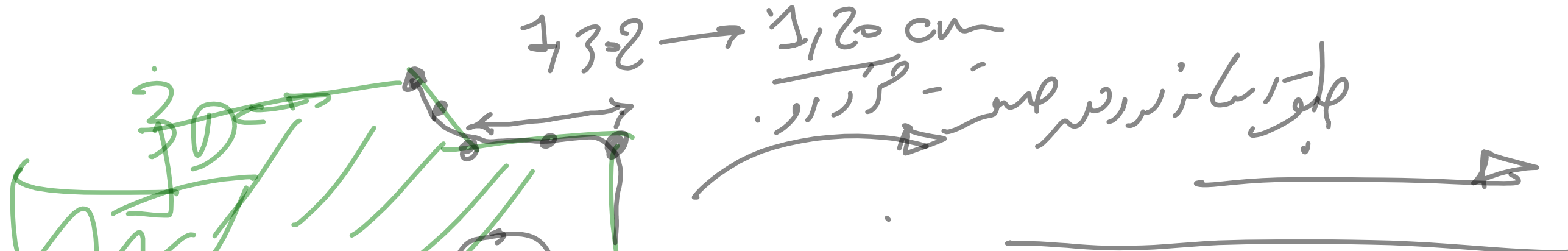
Figure 6. Keypoint detection results on COCO test using Mask R-CNN (ResNet-50-FPN), with person segmentation masks predicted from the same model. This model has a keypoint AP of 63.1 and runs at 5 fps.

شماره ۱۰۰



فوتبالیست





✓
 key point
 segment
 tracking!



Figure 7. Mask R-CNN results on Cityscapes test (32.0 AP). The bottom-right image shows a failure prediction.



30kp



(A) → 30kp

11

36



1392

3163



α.

	backbone	AP^{bb}	AP_{50}^{bb}	AP_{75}^{bb}	AP_S^{bb}	AP_M^{bb}	AP_L^{bb}
Faster R-CNN+++ [19]	ResNet-101-C4	34.9	55.7	37.4	15.6	38.7	50.9
Faster R-CNN w FPN [27]	ResNet-101-FPN	36.2	59.1	39.0	18.2	39.0	48.2
Faster R-CNN by G-RMI [21]	Inception-ResNet-v2 [37]	34.7	55.5	36.7	13.5	38.1	52.0
Faster R-CNN w TDM [36]	Inception-ResNet-v2-TDM	36.8	57.7	39.2	16.2	39.8	52.1
Faster R-CNN, RoIAlign	ResNet-101-FPN	37.3	59.6	40.3	19.8	40.2	48.8
Mask R-CNN	ResNet-101-FPN	38.2	60.3	41.7	20.1	41.1	50.2
Mask R-CNN	ResNeXt-101-FPN	39.8	62.3	43.4	22.1	43.2	51.2

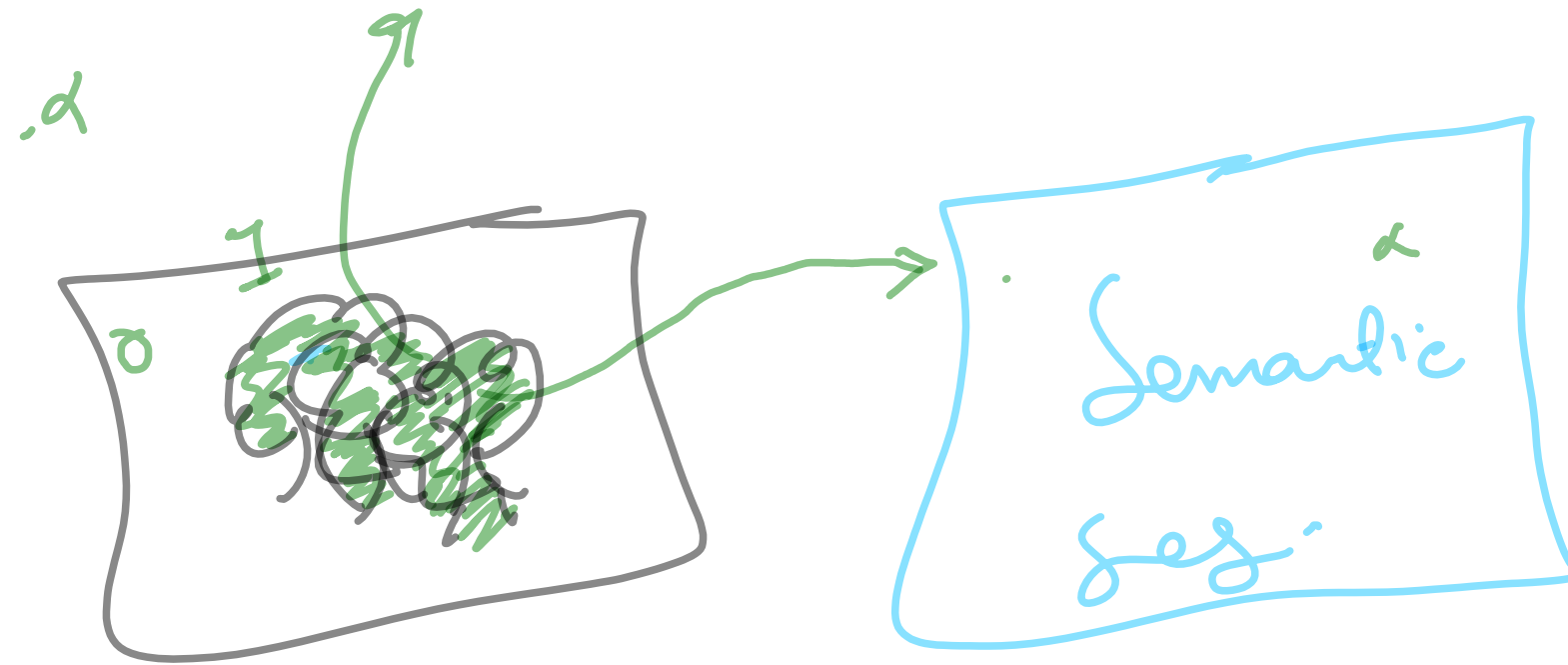
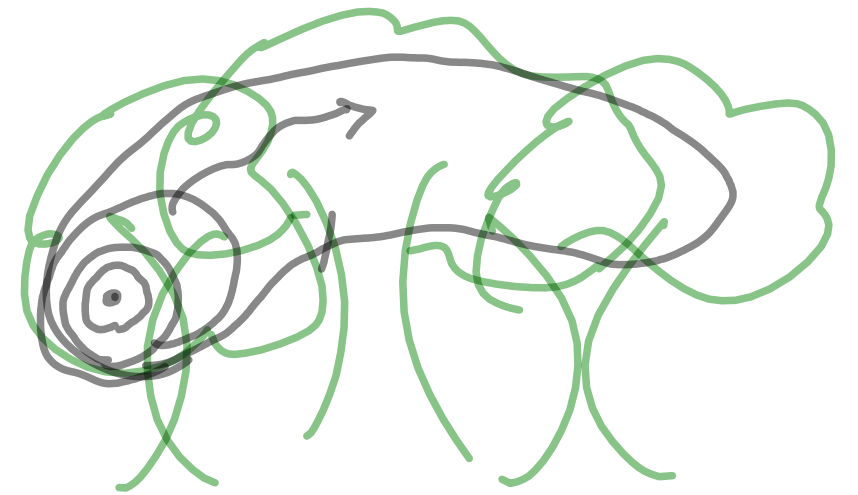
Table 3. **Object detection** *single-model* results (bounding box AP), vs. state-of-the-art on `test-dev`. Mask R-CNN using ResNet-101-FPN outperforms the base variants of all previous state-of-the-art models (the mask output is ignored in these experiments). The gains of Mask R-CNN over [27] come from using RoIAlign (+1.1 AP^{bb}), multitask training (+0.9 AP^{bb}), and ResNeXt-101 (+1.6 AP^{bb}).

	backbone	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
MNC [10]	ResNet-101-C4	24.6	44.3	24.8	4.7	25.9	43.6
FCIS [26] +OHEM	ResNet-101-C5-dilated	29.2	49.5	-	7.1	31.3	50.0
FCIS+++ [26] +OHEM	ResNet-101-C5-dilated	33.6	54.5	-	-	-	-
Mask R-CNN	ResNet-101-C4	33.1	54.9	34.8	12.1	35.6	51.1
Mask R-CNN	ResNet-101-FPN	35.7	58.0	37.8	15.5	38.1	52.4
Mask R-CNN	ResNeXt-101-FPN	37.1	60.0	39.4	16.9	39.9	53.5

Table 1. **Instance segmentation** *mask* AP on COCO `test-dev`. MNC [10] and FCIS [26] are the winners of the COCO 2015 and 2016 segmentation challenges, respectively. Without bells and whistles, Mask R-CNN outperforms the more complex FCIS+++, which includes multi-scale train/test, horizontal flip test, and OHEM [35]. All entries are *single-model* results.



object!



α

α

Instance.

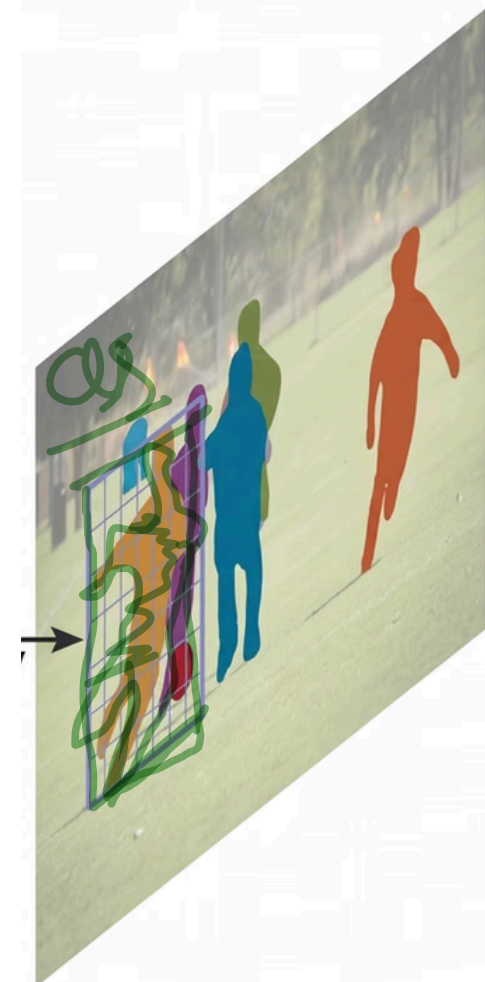
Segmentation



CLS

BBOX

MASK



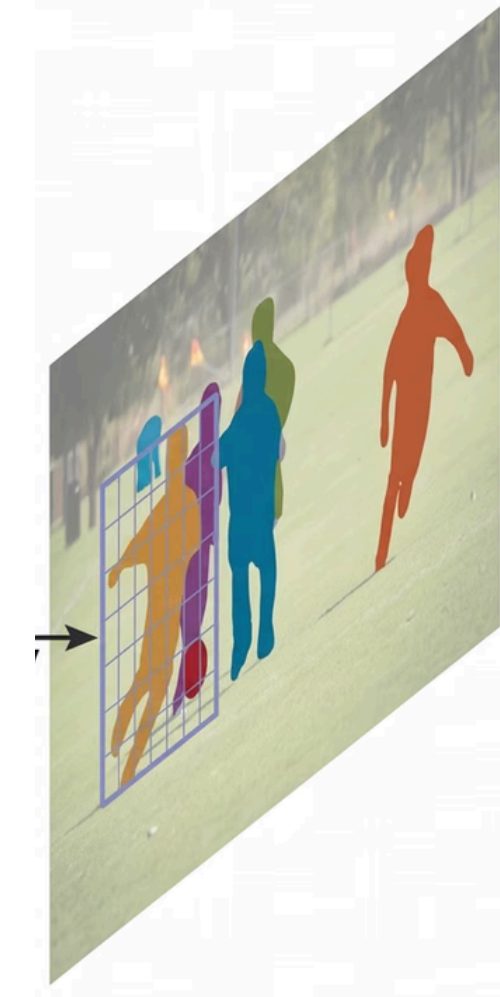
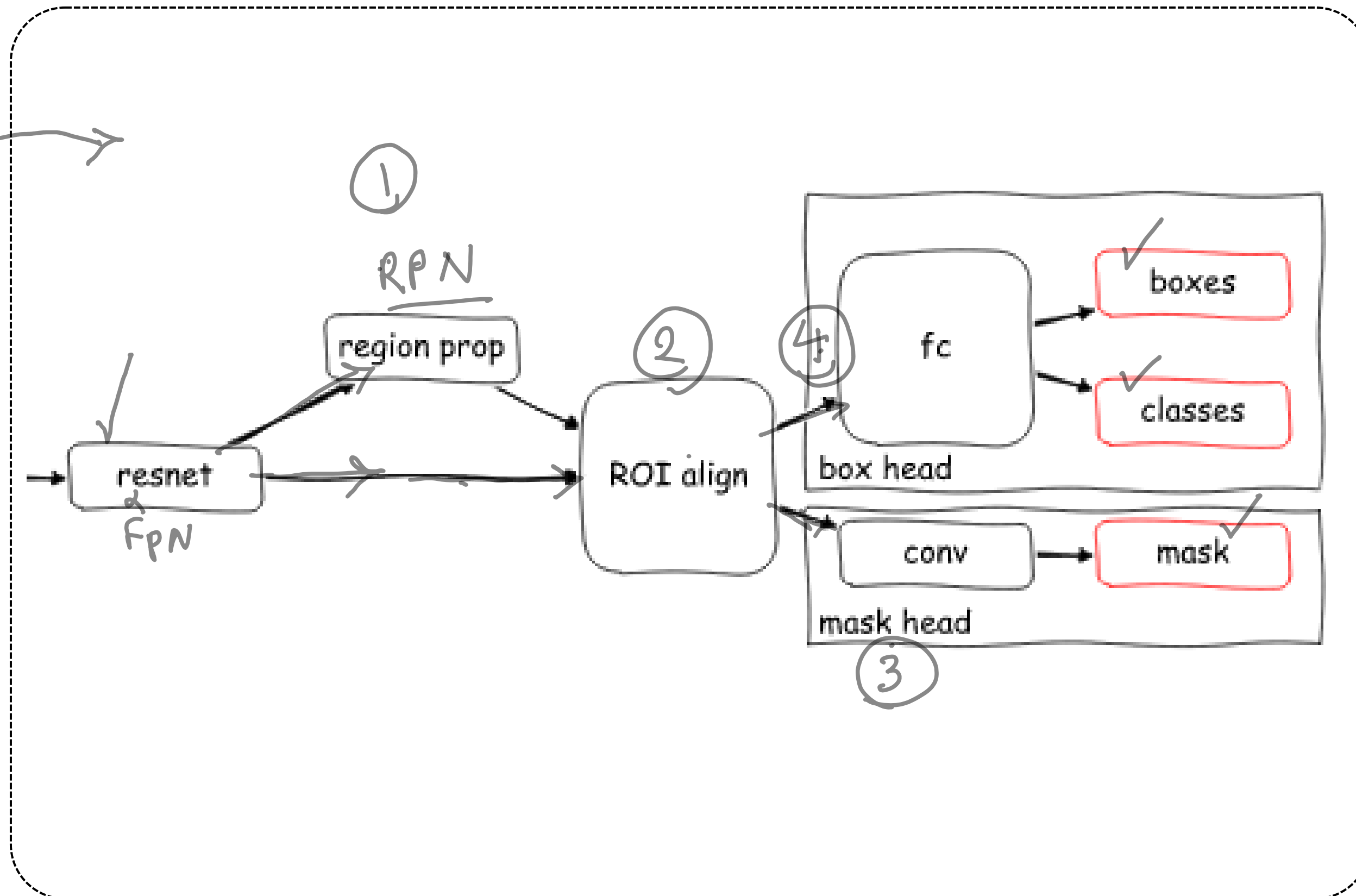
Instances!

mask!

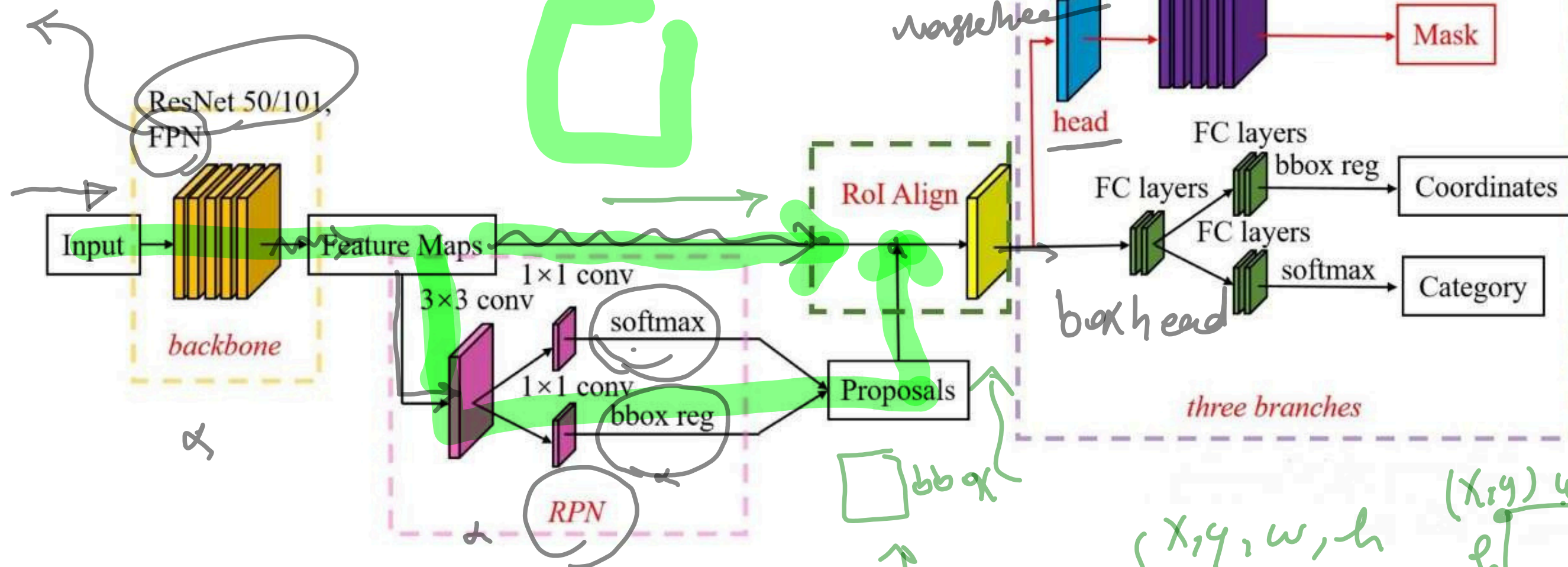
- Loss function for each sampled RoI

$$L = L_{cls} + L_{box} + L_{mask}$$

Mask R-CNN



pyramid
feature proposal net!

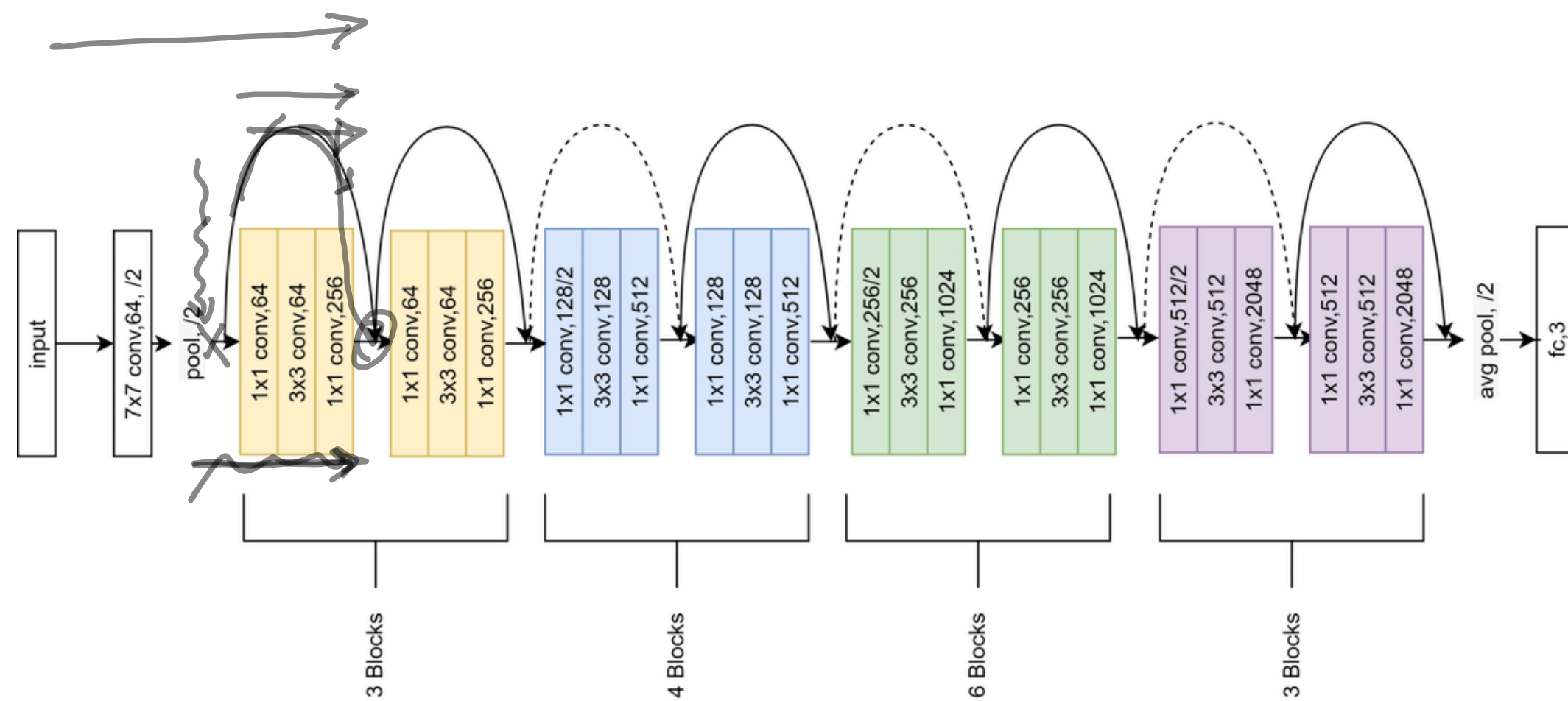
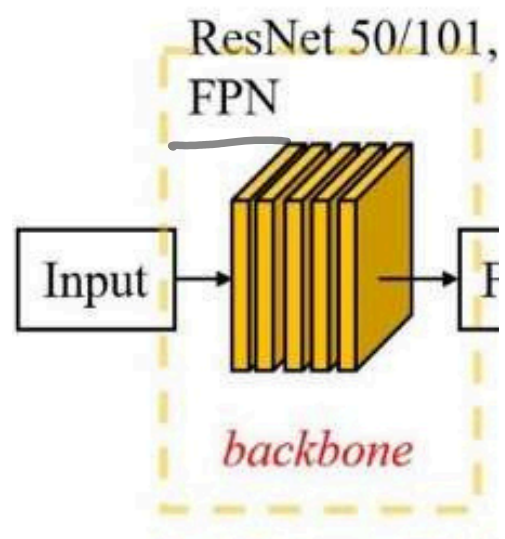


bbox

$\begin{cases} x, y, w, h \\ os, cls \end{cases}$

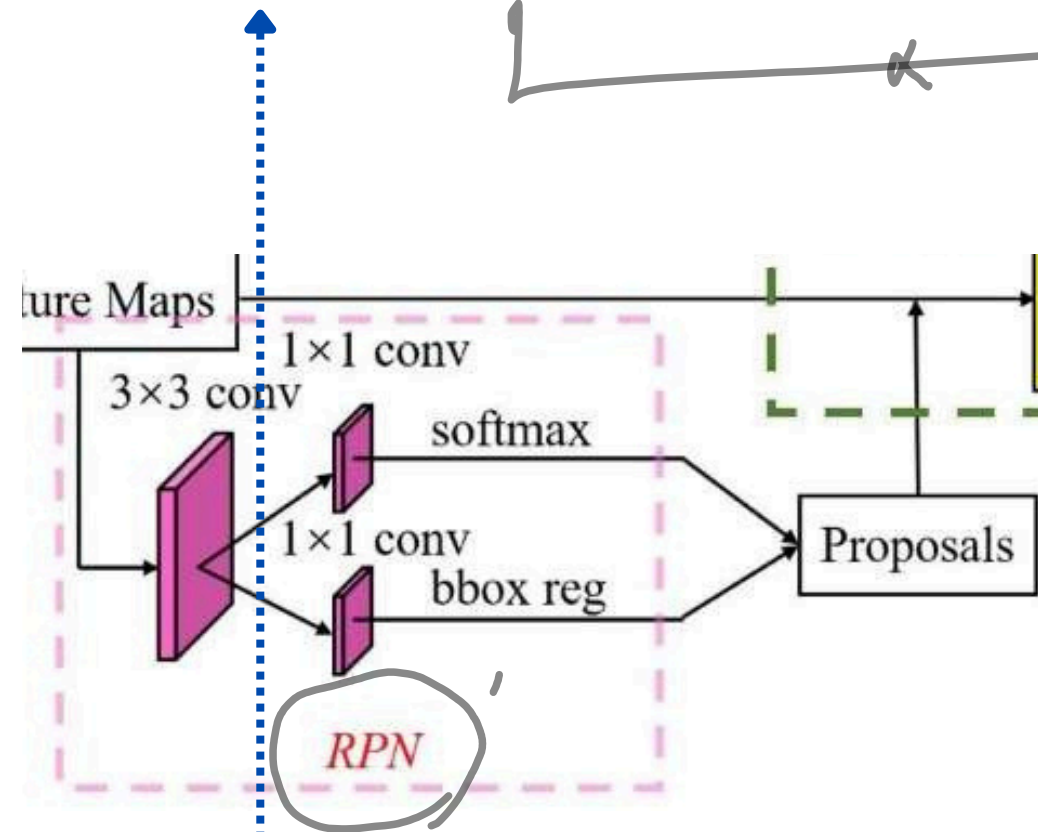
$(x, y) \begin{matrix} w \\ h \end{matrix} \rightarrow os, cls$

$[x, y, w, h, os, cls]$



ResNet-50 Layered architecture

Region proposal net.

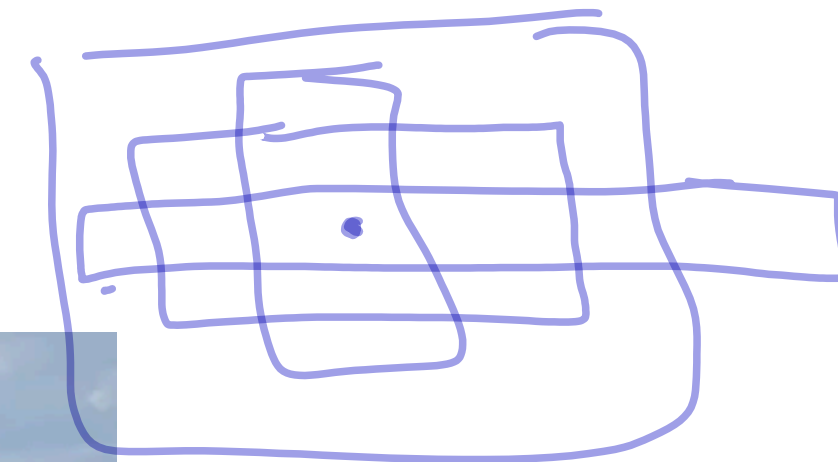


• وظیفه دارد نواحی ای از تصویر را پیدا کند که احتمالاً شامل یک شیء هستند.

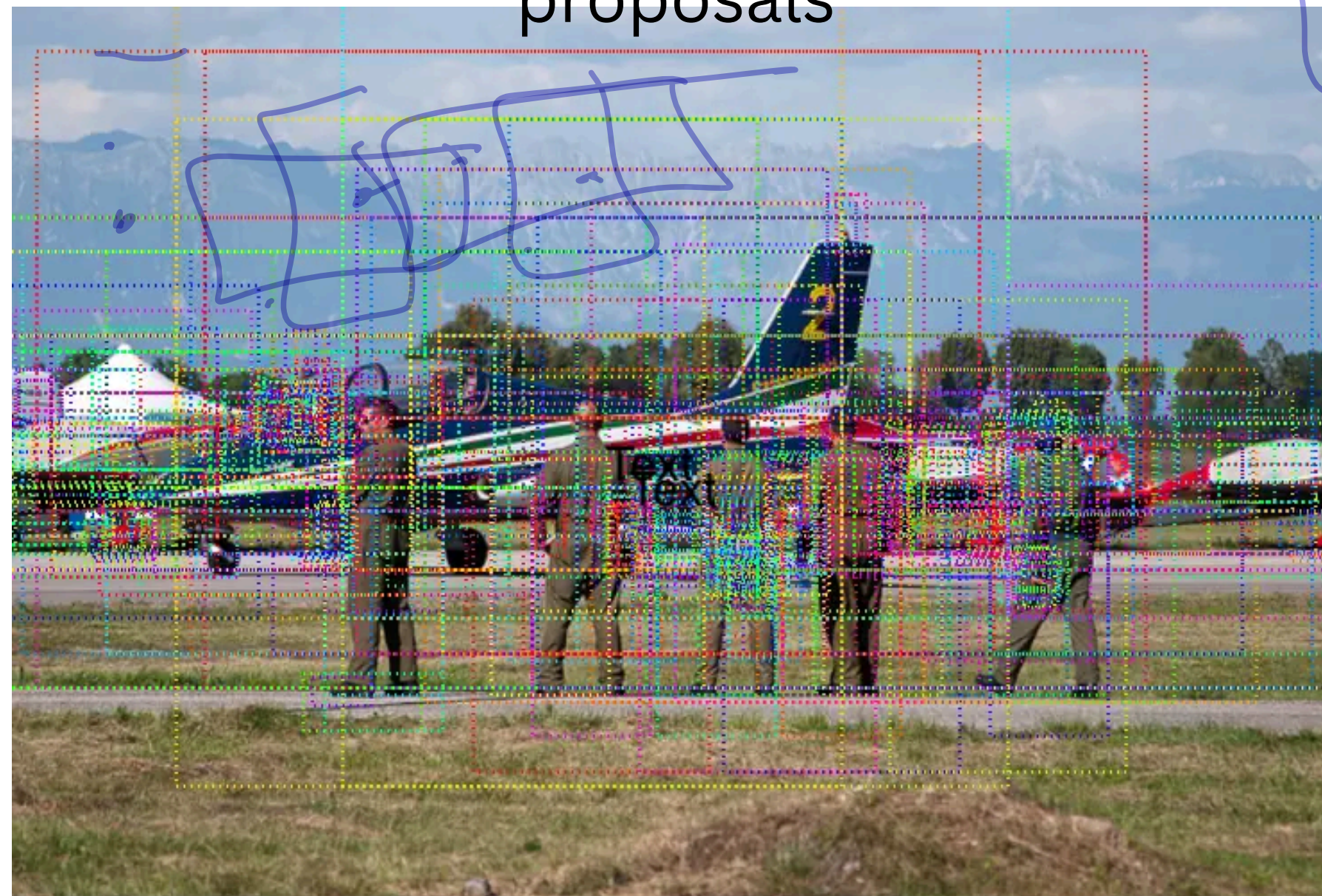
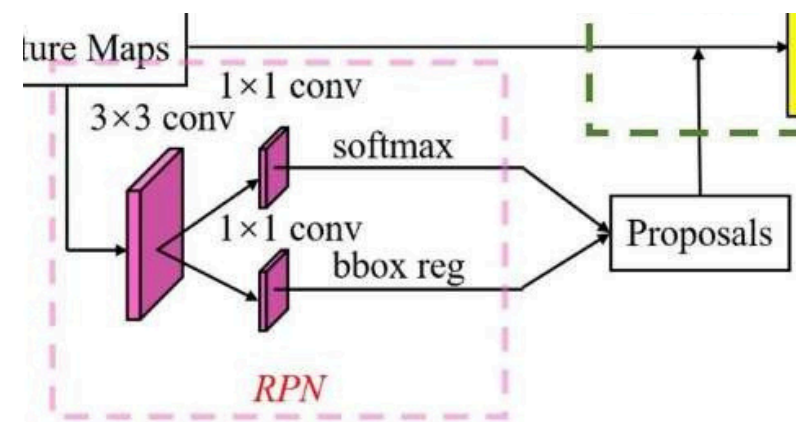
• خروجی RPN تعدادی کادر پیشنهادی یا Region Proposal است که در مراحل بعدی Mask R-CNN بررسی می شوند.

Intermediate Feature Map

First Regions proposals



RPN

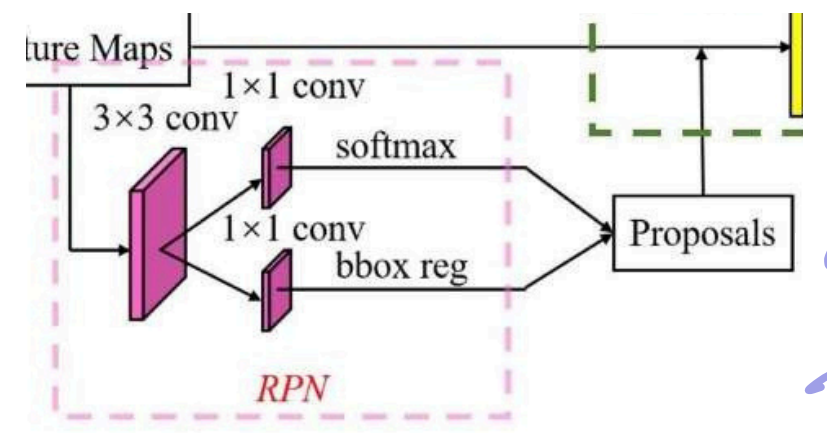


• RPN روی تمام این Feature Map ها حرکت می کند و در هر نقطه تعدادی کادر اولیه به نام Anchor قرار می دهد.

$$\begin{cases} w-a = 9px \\ h-a = 12px \end{cases} \rightarrow \begin{cases} w-a-r = w \times 9 \\ h-a-r = w \times 12 \end{cases}$$

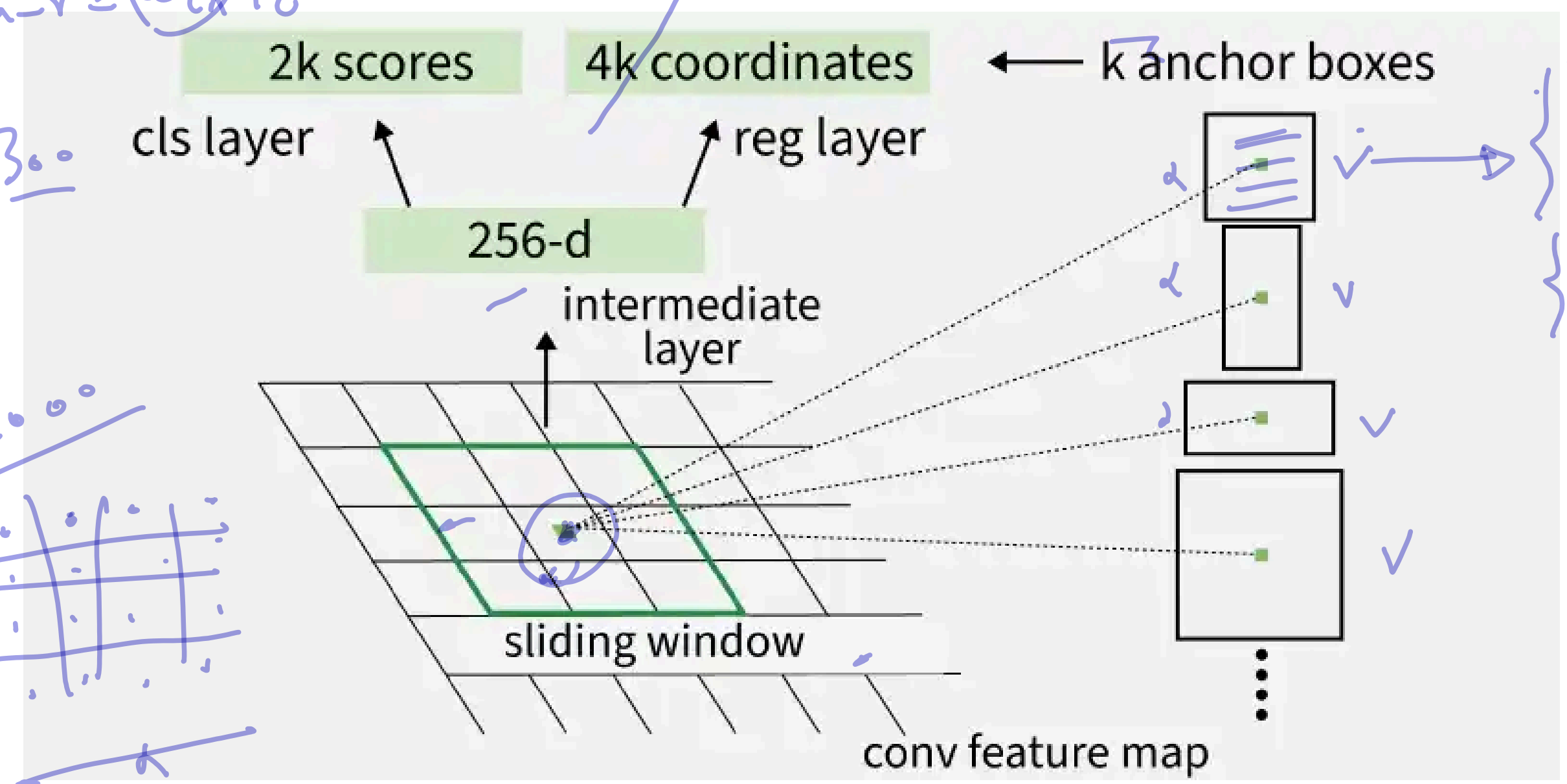
RPN

$$K = 300$$



$K \rightarrow$ Anchor

$$\begin{cases} 4K \rightarrow \text{coord.} \\ 2K \rightarrow \text{os} \end{cases} \quad \therefore \textcircled{n}$$

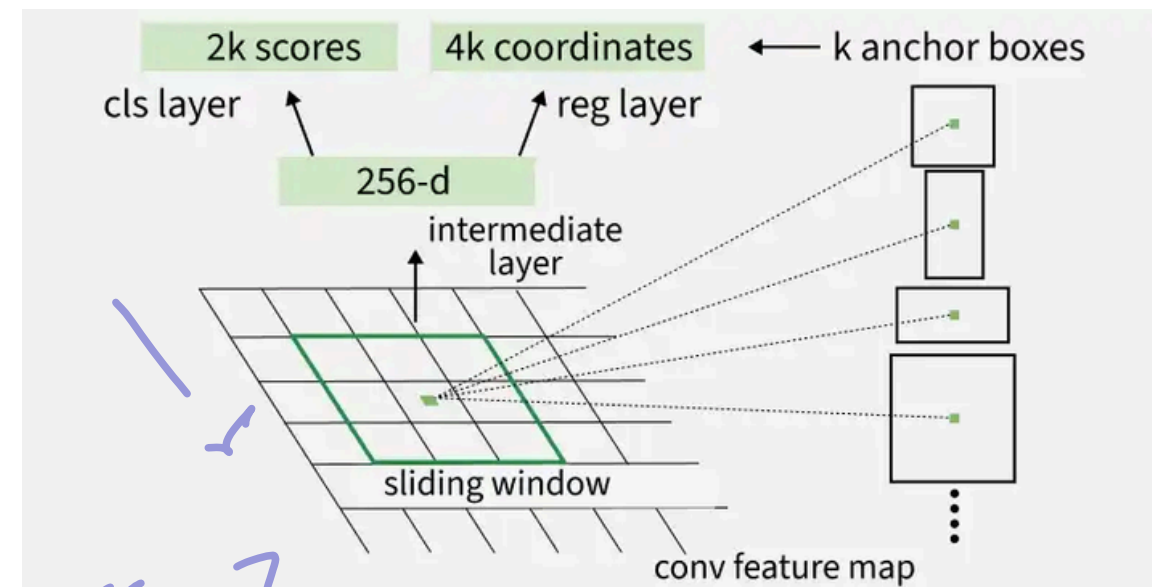
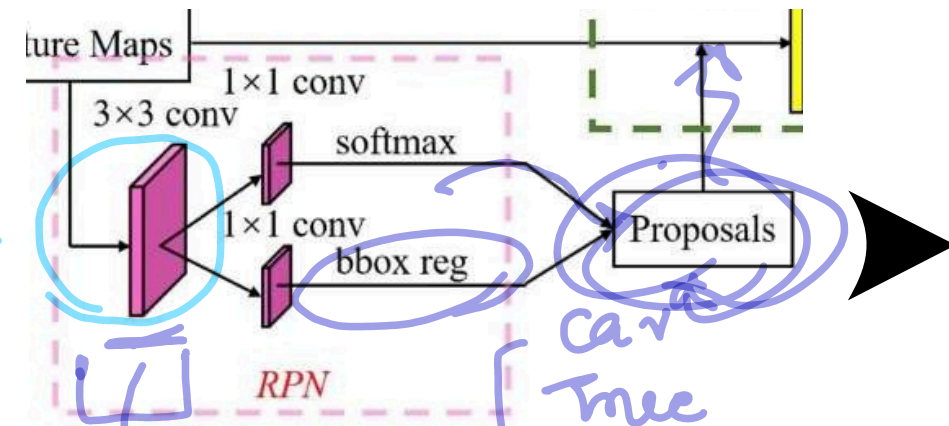


$$\begin{cases} w, h \\ n, y \end{cases} \textcircled{4}$$

$$\begin{cases} as = ? , 4 \\ cls = ? + reg \end{cases}$$

• همه این Anchor ها یک مرکز مشترک دارند، اما اندازه و نسبت ابعادشان متفاوت است.

RPN



$$P2 \rightarrow 32 \times 32$$

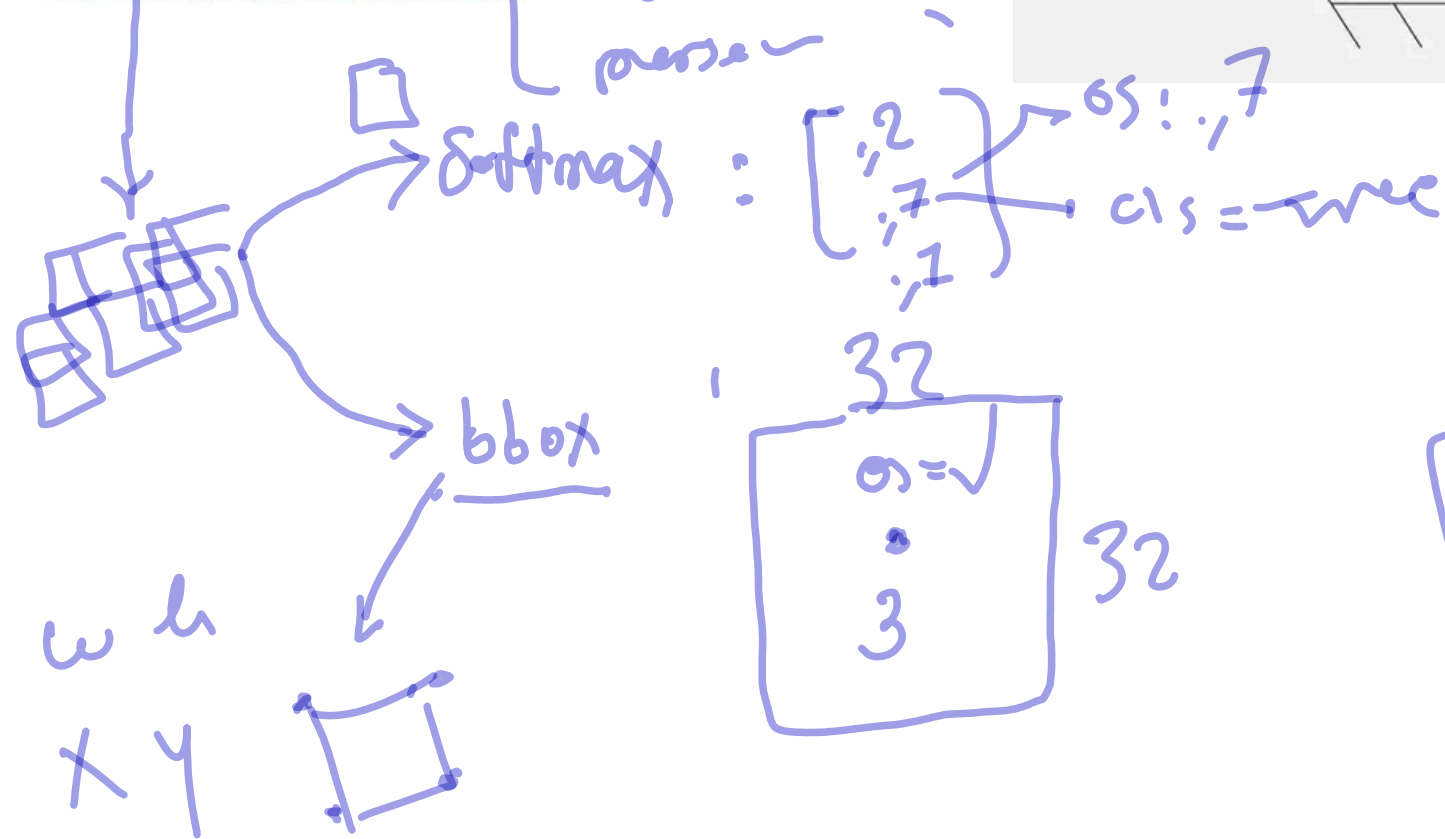
$$P3 \rightarrow 64 \times 64$$

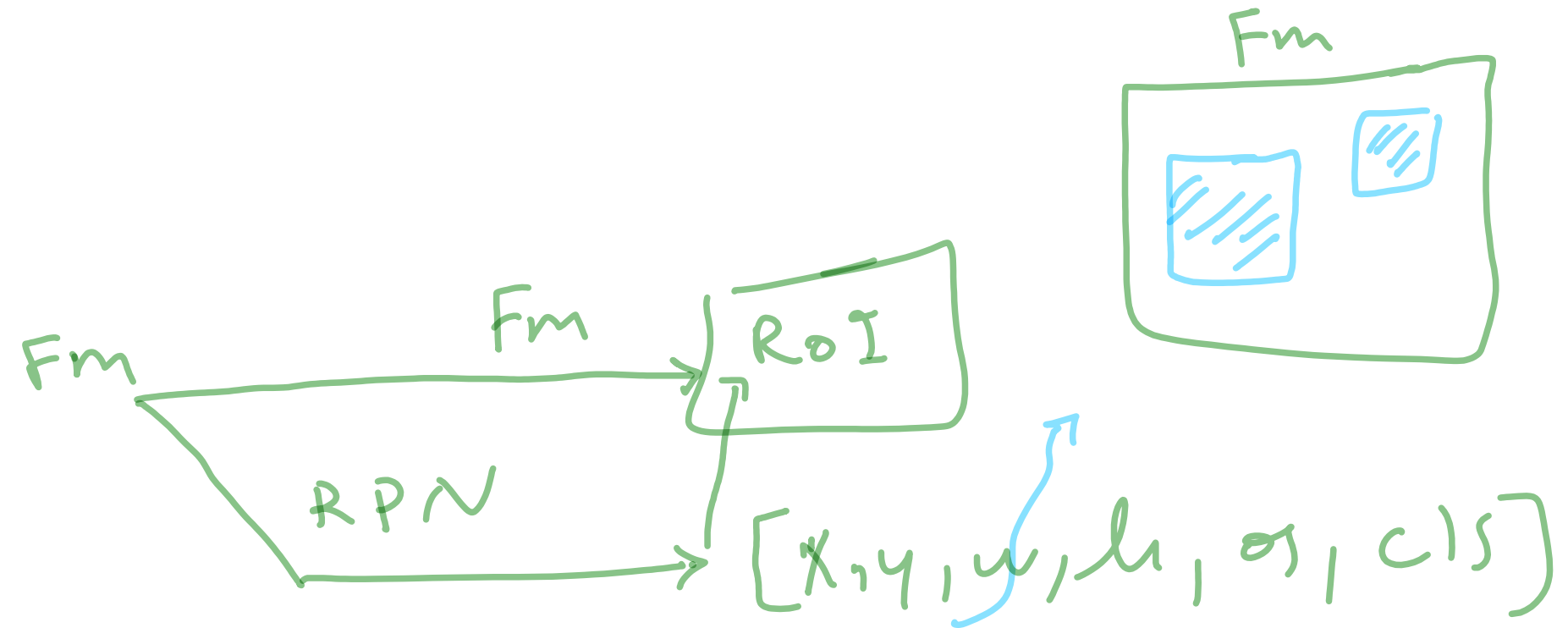
$$P4 \rightarrow 128 \times 128$$

$$P5 \rightarrow 256 \times 256$$

$$P6 \rightarrow 512 \times 512$$

ممکن است Anchor های پایه چنین باشند



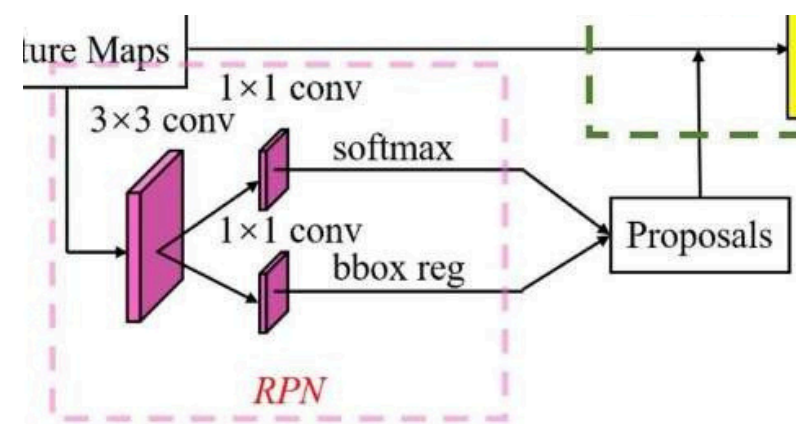


$$\Delta x, \Delta y, \Delta w, \Delta h$$

$$\alpha$$

First Regions proposals

RPN

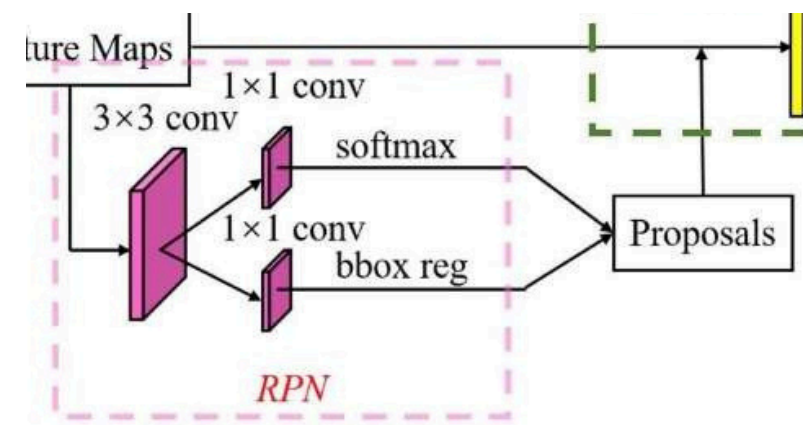


• RPN روی تمام این Feature Map ها حرکت می کند و در هر نقطه تعدادی کادر اولیه به نام Anchor قرار می دهد.

• آیا این Anchor شامل شیء است؟

• مختصات Anchor چقدر باید تغییر کند تا بهتر روی شیء قرار بگیرد؟

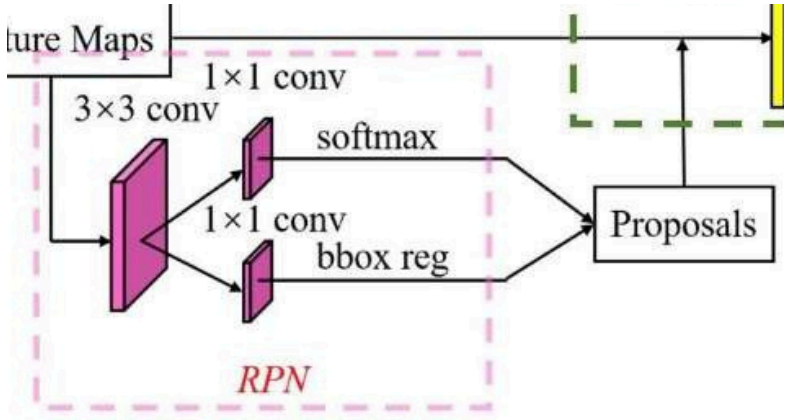
شاخه Objectness



آیا این شامل شیء است؟

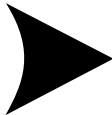


شاخه Objectness

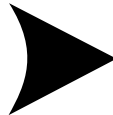


آیا این Anchor شامل شیء است؟

Objectness Scores



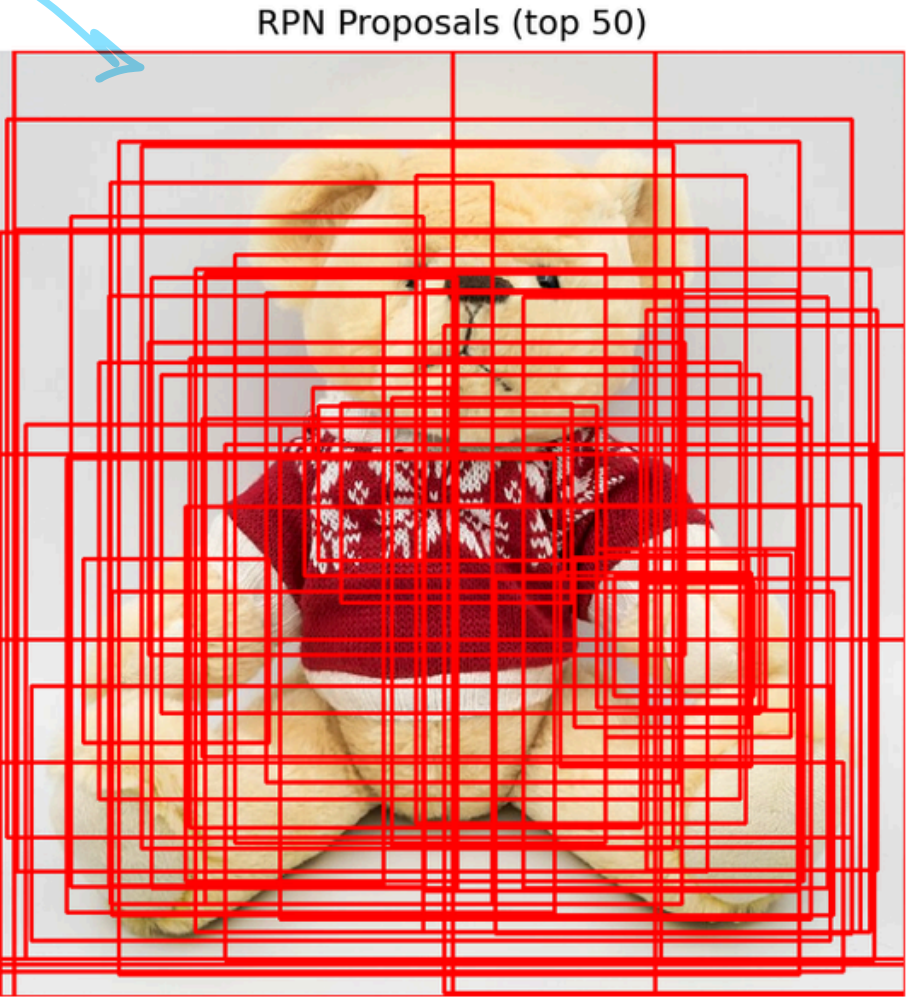
Foreground: شامل شیء
Background: فاقد شیء



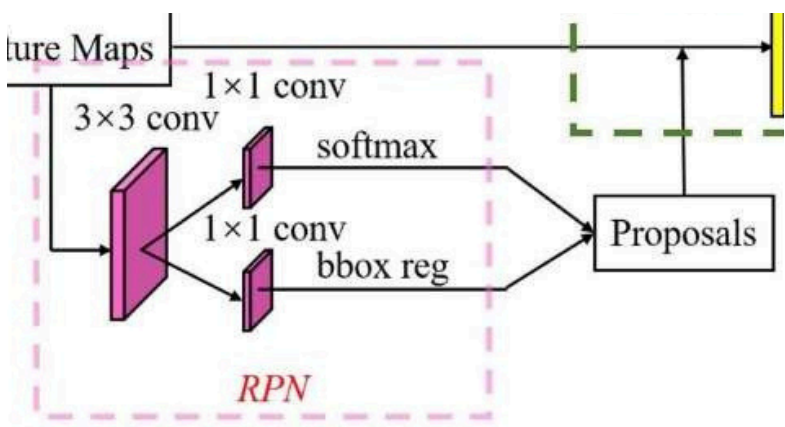
Anchor 1 → 0.94
Anchor 2 → 0.62
Anchor 3 → 0.08



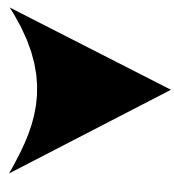
• احتمال وجود شیء



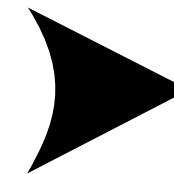
شاخه Bounding Box Regression



Intermediate Features



1 × 1 Convolution



Bounding Box Deltas

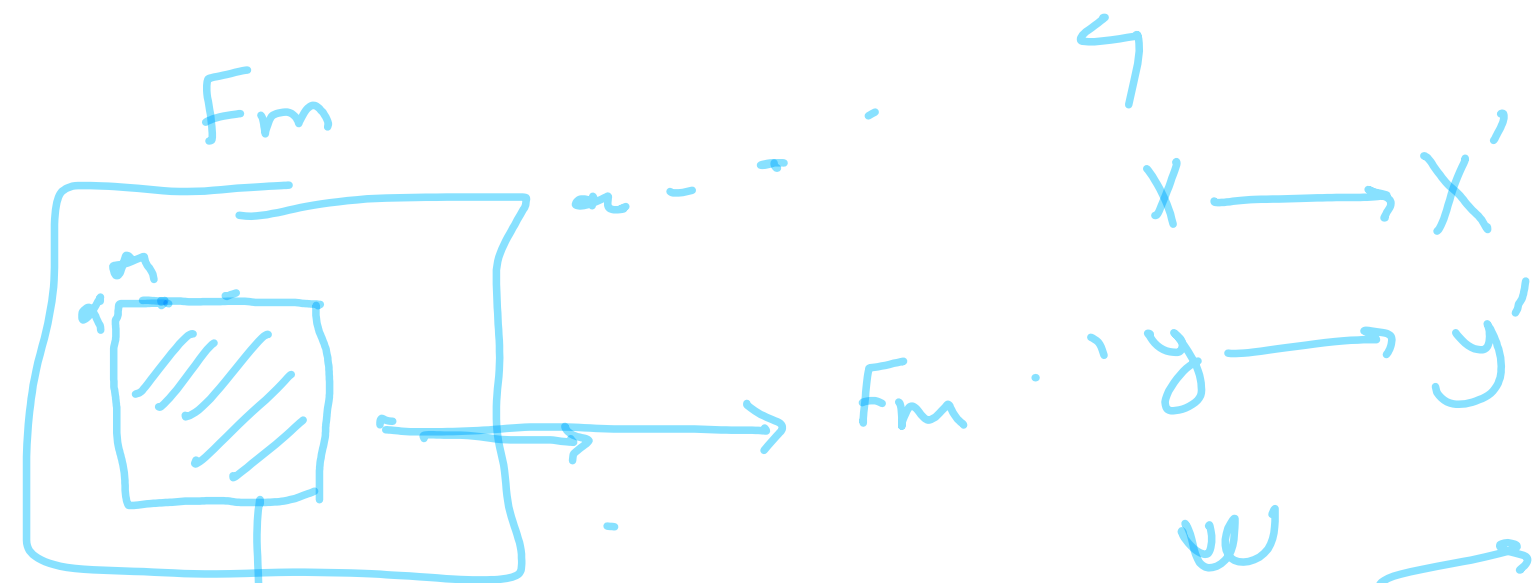
- tx: جابه‌جایی افقی مرکز کادر
- ty: جابه‌جایی عمودی مرکز کادر
- tw: تغییر عرض کادر
- th: تغییر ارتفاع کادر

مختصات Anchor چقدر باید تغییر کند تا بهتر روی شیء قرار بگیرد؟

7

$$\begin{aligned}x &= t_x w_a + x_a \\ y &= t_y h_a + y_a \\ w &= e^{t_w} w_a \\ h &= e^{t_h} h_a\end{aligned}$$

• مختصات جدید



RPN

Conv / ...

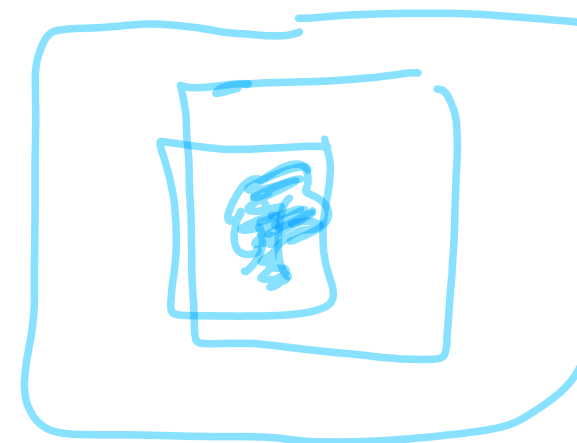
new

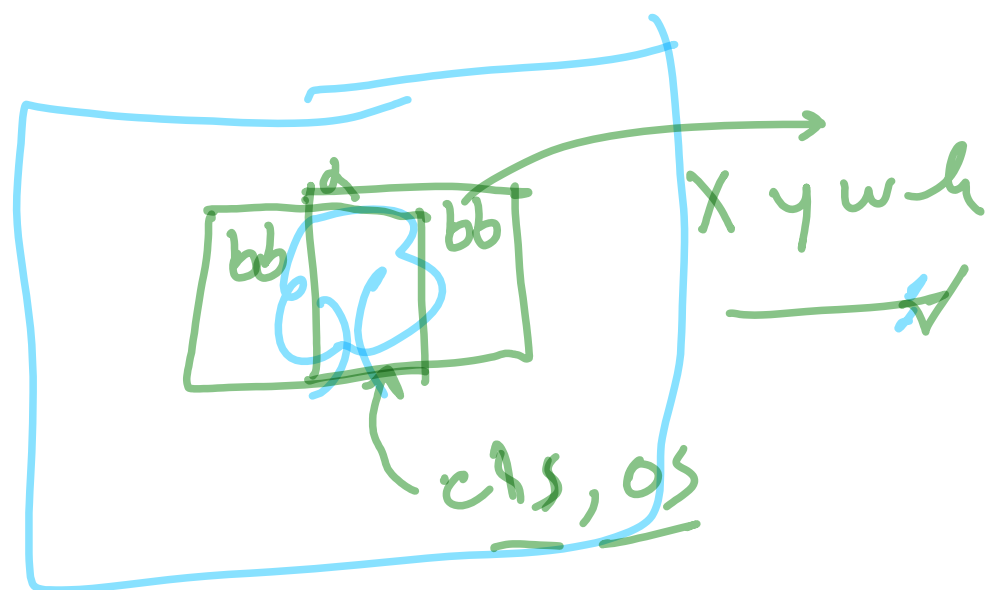
Δx
 Δy
 Δw
 Δh

x
 y
 w
 h

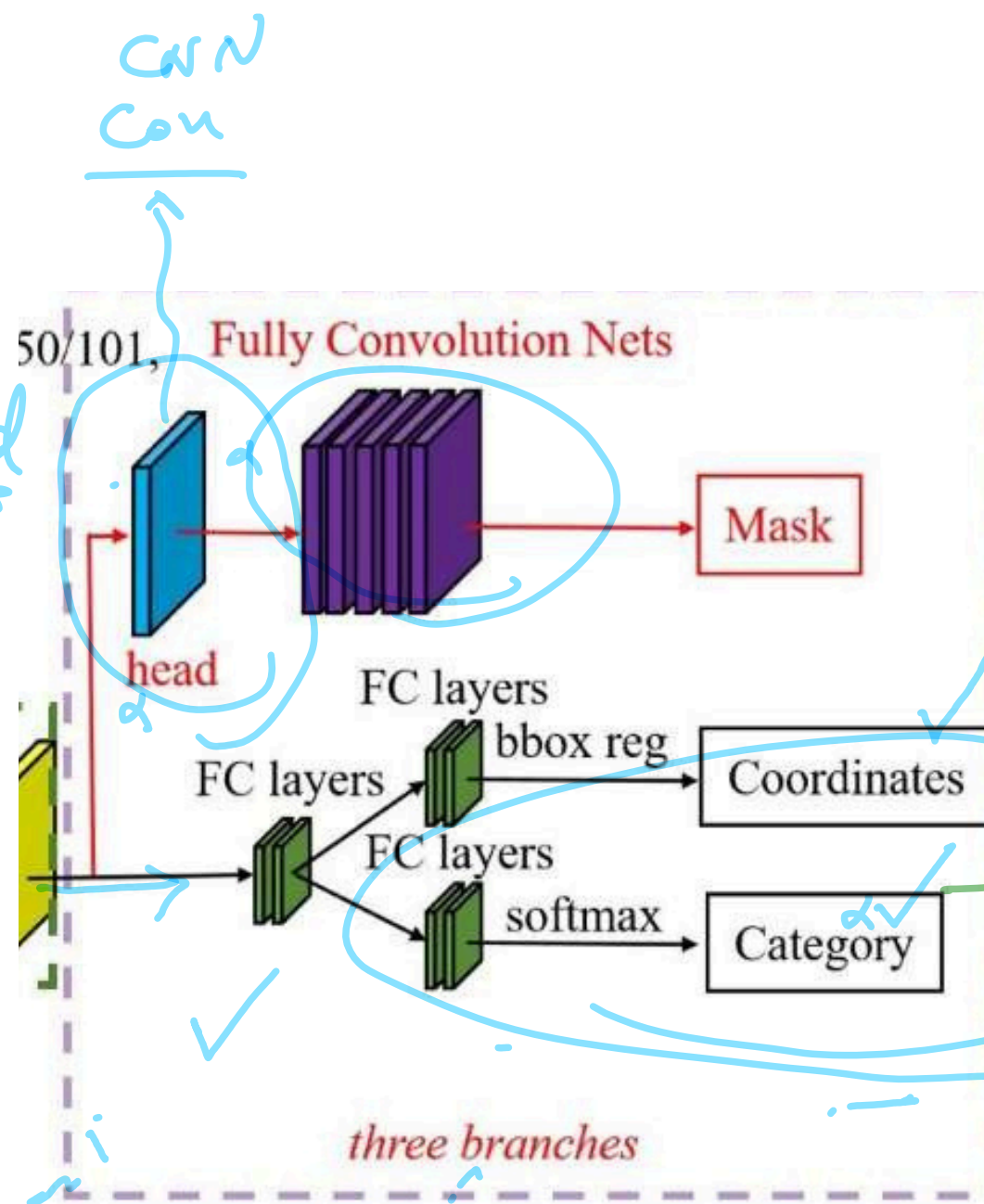


$F_n / (x', y', w', h', \theta, cls)$

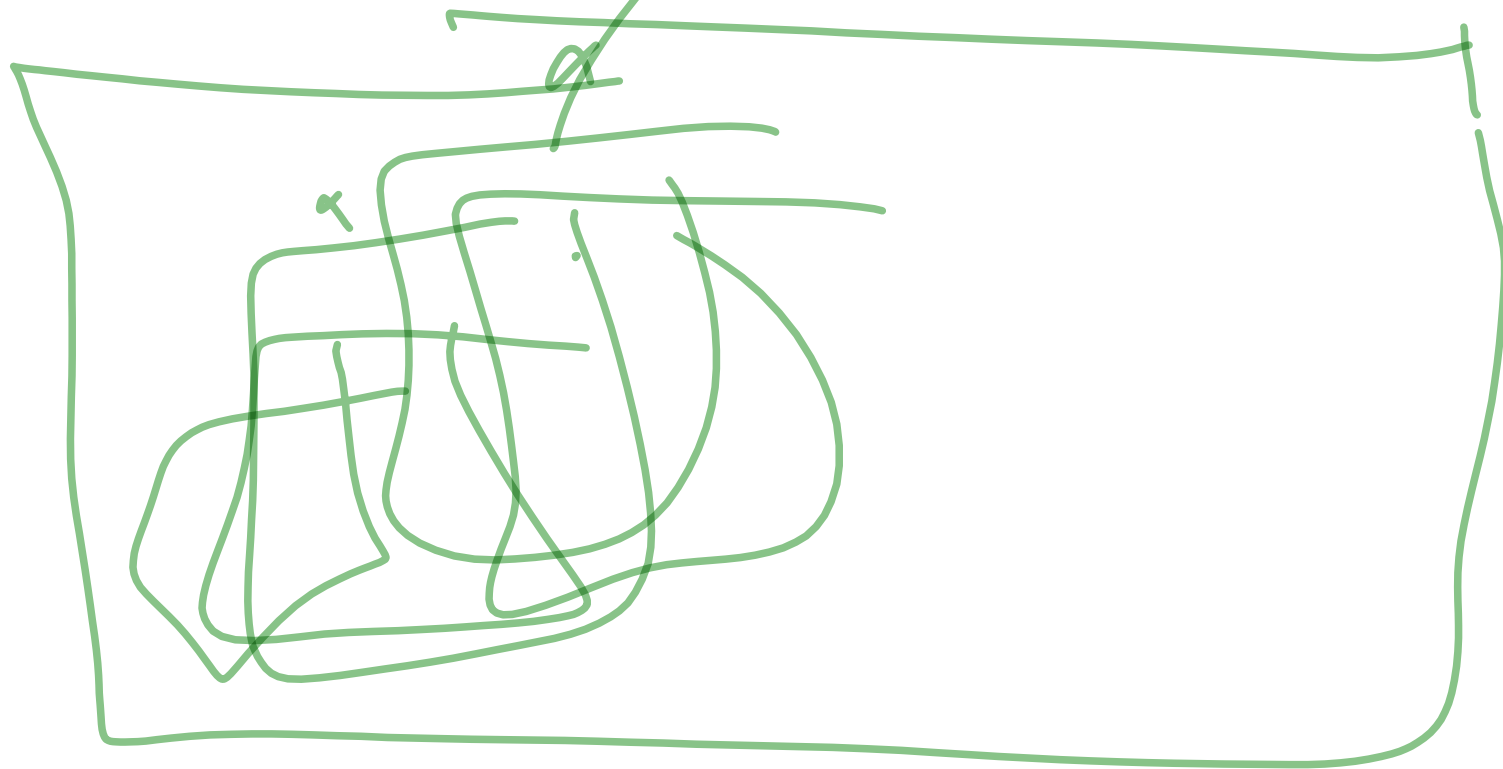
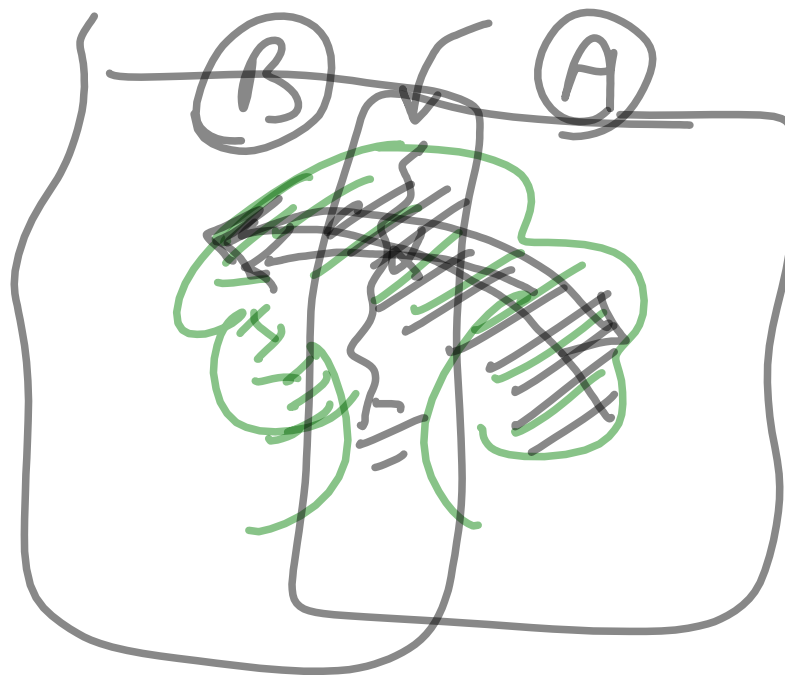




mask head



mask ✓
Coordinates ✓
cls ✓



Hele
